Lesson 1.2.4 — Functions

LEARNING FOCUS

1. **1.2.4** — I can distinguish functions from relationships that are not functions.
2. **1.2.5** — I can identify the domain and range of a relation.
3. **2.1.1** — I can connect linear growth in tables, graphs, and equations.

EXAMPLES & YOU TRY IT

Example 1 — work together

Does the set $\{(1,3),\ (2,4),\ (1,5),\ (3,6)\}$ represent a function? Explain using the inputs.

You Try It 1 — independently

Does $\{(1,2),\ (2,4),\ (3,2),\ (4,4)\}$ represent a function? Explain.

NOTES

Math Notes

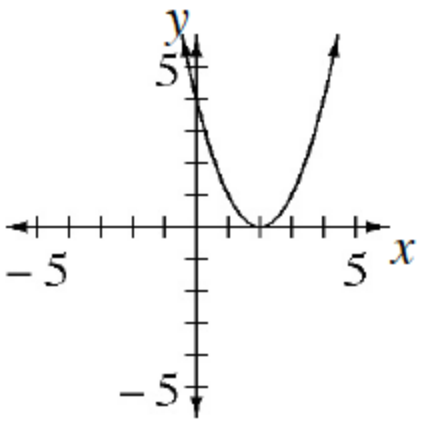
METHODS AND MEANINGS
MATH NOTES

Functions

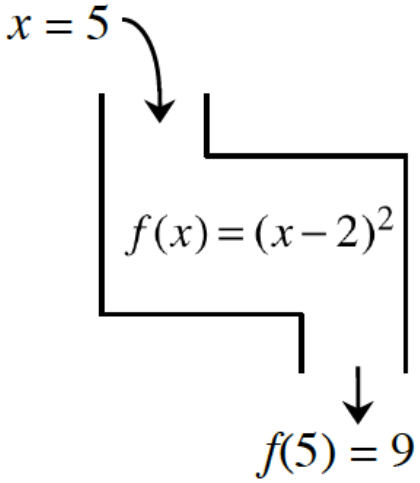
A relationship between inputs and outputs is a function if there is exactly one output for each input. We often write a function as $y =$ some expression involving x, where x is the input and y is the output. The following is an example of a function.

$y = (x - 2)^2$

x	-2	-1	0	1	2	3	4	5
y	16	9	4	1	0	1	4	9



In the example above the value of y depends on x, so *y* is also called the dependent variable and x is called the independent variable.



Another way to write a function is with the notation "$f(x) =$" instead of "$y =$". The function named "f" has output $f(x)$. The input is x.

Math Notes (continued)

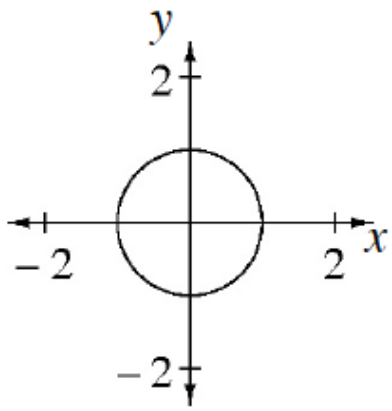
In the example at right, $f(5) = 9$. The input is 5 and the output is 9. You read this as, "f of 5 equals 9."

The set of all inputs for which there is an output is called the domain. The set of all possible outputs is called the range. In the example above, notice that you can input any x-value into the equation and get an output. The domain of this function is "all real numbers" because any number can be an input. But the outputs are all greater than or equal to zero. The range is $y \geq 0$.

$x^2 + y^2 = 1$ is not a function because there are two y-values (outputs) for some x-values, as shown below.

$x^2 + y^2 = 1$

x	−1	0	0	1
y	0	−1	1	0



Example 2 — work together

Find the domain and range of $\{(-2, 1), (0, 4), (3, 1), (5, 7)\}$. Write each as a set without repeats.

You Try It 2 — independently

Find domain and range of $\{(-3, 2), (1, 6), (4, 2), (8, 9)\}$.

Example 3 — work together

Given $y = 2x + 1$, make a table for x = 0, 1, 2, and 3. Plot the points and explain the growth pattern.

You Try It 3 — independently

Now use $y = 3x + 2$. Make a table for x = 0, 1, 2, and 3, graph the points, and describe the growth.

REVIEW & PREVIEW

1-66

If $g(x) = \sqrt{x-7}$, find $g(8)$, $g(32)$, and $g(80)$.

1-67

Solve each equation below. Check each solution.

a. $6 - x - 3 = 10$

b. $100x + 300 = 200$

c. $\frac{1}{3}x + 4 = x - 2$

d. $36 - 2x = -x + 2$

1-68

Find $f(-4)$ for each function below.

a. $f(x) = |x - 3|$

b. $f(x) = -5\,|x|$

c. $f(x) = |x + 1|$

d. $f(x) = |x + 3| - 6$

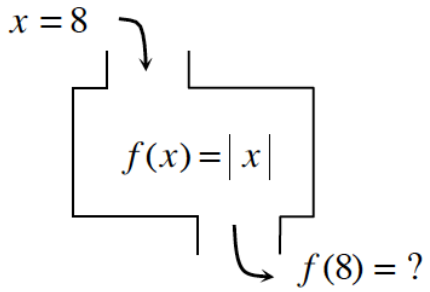
1-69

Graph and fully describe the function $f(x) = -x^2 + 3$. Graph values of x from -3 to 3. .

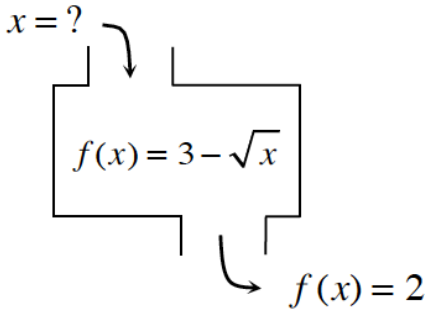
1-70

Find the corresponding inputs or outputs for the following functions. If there is no solution, explain why not. Be careful: In some cases, there may be no solution or more than one possible solution.

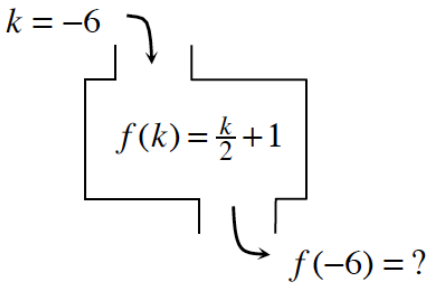
a.



b.



c.



d.

